Can Daily Lifestyle Factors Predict Stress Levels?

Neural Network Classification Using Mental Health and Lifestyle Data

Introduction

Stress and mental health are influenced by many lifestyle and behavioral factors. This project explores whether daily habits such as sleep, exercise, screen time, and workload can be used to predict stress classifications using baseline predictors and neural network models.

The goal was to compare simple rule-based approaches against neural networks while studying feature importance, model complexity, and training performance.

Workflow

Raw Dataset
↓
EDA + Correlation Analysis
↓
Stress Classification
↓
Normalization
↓
Baseline Predictors
↓
Neural Network Models
↓
Evaluation & Feature Importance

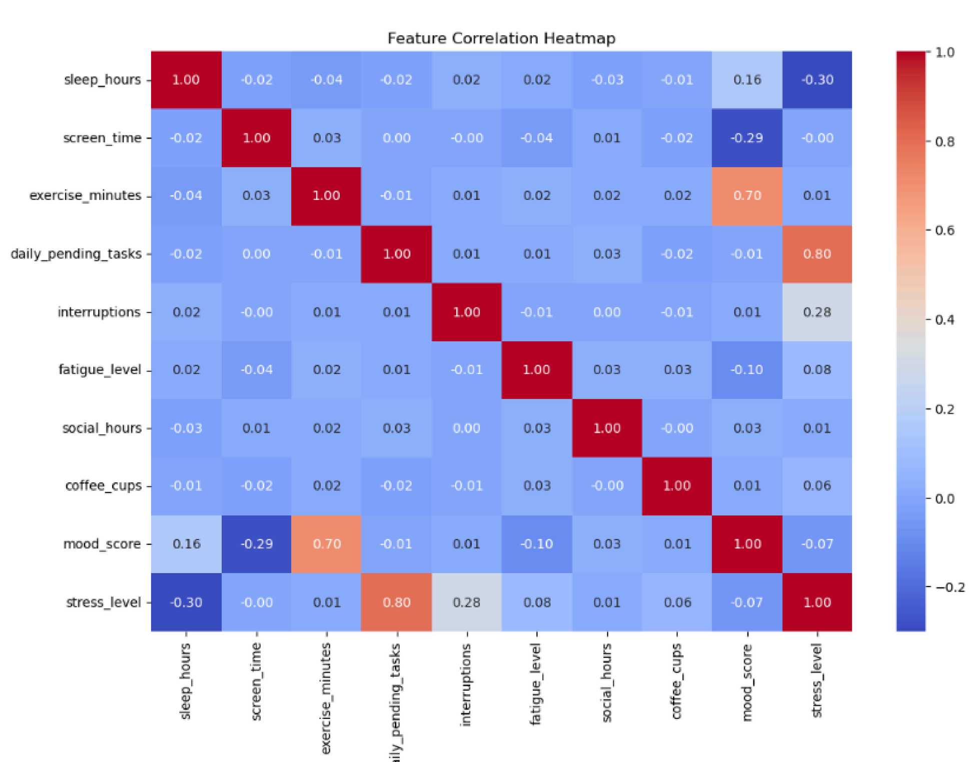
Sleep Hours – total hours slept
Screen Time – daily time spent on digital devices
Exercise Minutes – physical activity per day
Daily Pending Tasks – unfinished work items or responsibilities
Interruptions – number of daily interruptions
Fatigue Level – subjective fatigue rating
Social Hours – time spent interacting with family, friends, or coworkers
Coffee Cups – caffeine intake
Diet Quality – categorical rating of meals: poor, average, or good
Weather – categorical: sunny, cloudy, rainy, snowy
Mood Score – daily mood rating (1–10)
Stress Level – daily stress rating (1–10)

Dataset

- Source: Kaggle synthetic mental health/lifestyle dataset
- ~2000 observations
- Tabular dataset containing numerical lifestyle and behavioral features

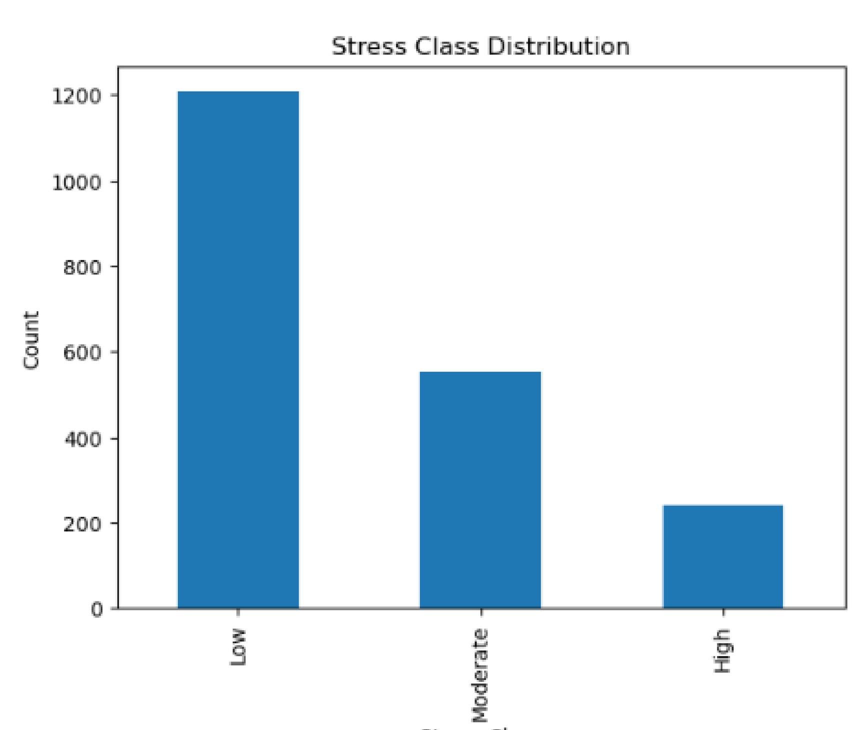
Exploratory Data Analysis

Initial exploratory analysis was performed using histograms, summary statistics, and correlation analysis. Correlation analysis showed that some features had much stronger relationships with stress levels than others.



Data Preparation

Before training, all numerical features were normalized to place them on similar scales. The original stress score was also converted into three stress classification categories (Low, Moderate, and High) using manually selected threshold values.



Baseline Predictors

Three progressively more advanced baseline predictors were created before building neural networks.

Predictor 1 — Weighted Additive Model

A simple additive scoring system using weighted feature values.

Predictor 2 — Rule-Based Threshold Model

A manually designed classification system using logical thresholds.

Predictor 3 — Hybrid Predictor

A combination of additive scoring and rule-based logic.

$$S = b + w_1x_1 + w_2x_2 - w_3x_3$$

S = predicted stress score

b = baseline stress

x_i = normalized feature values

w_i = feature weights / importance

Predictor	Type	Accuracy
Predictor 1	Weighted Additive	58.9%
Predictor 2	Rule-Based Threshold	63.4%
Predictor 3	Hybrid Rule-Weighted	80.0%

Neural Network Development

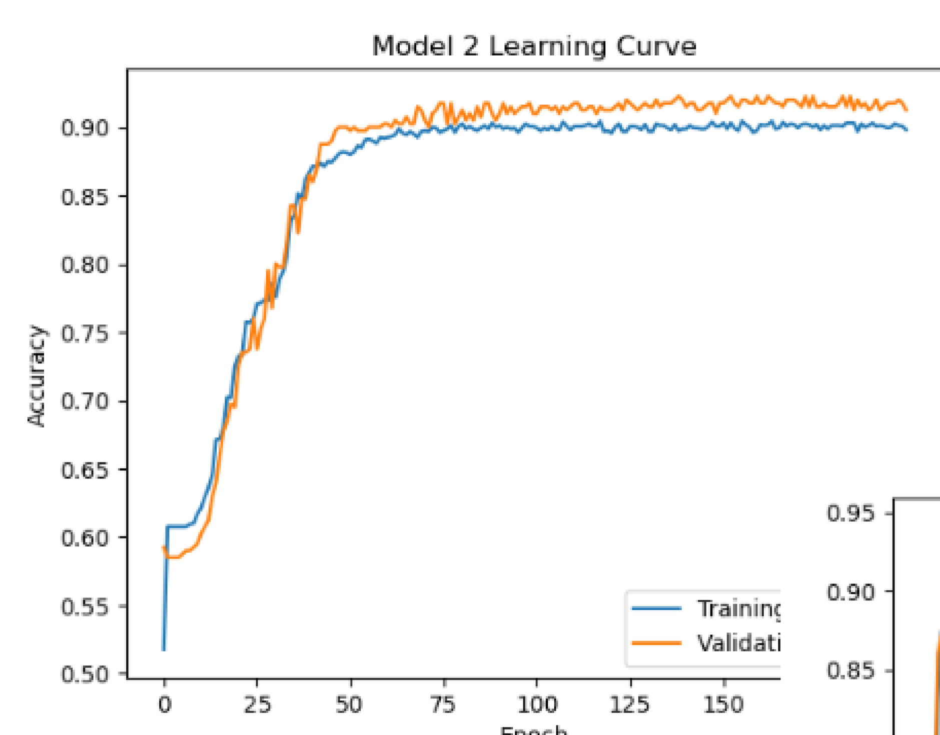
Multiple neural network architectures were tested to study model behavior and overfitting.

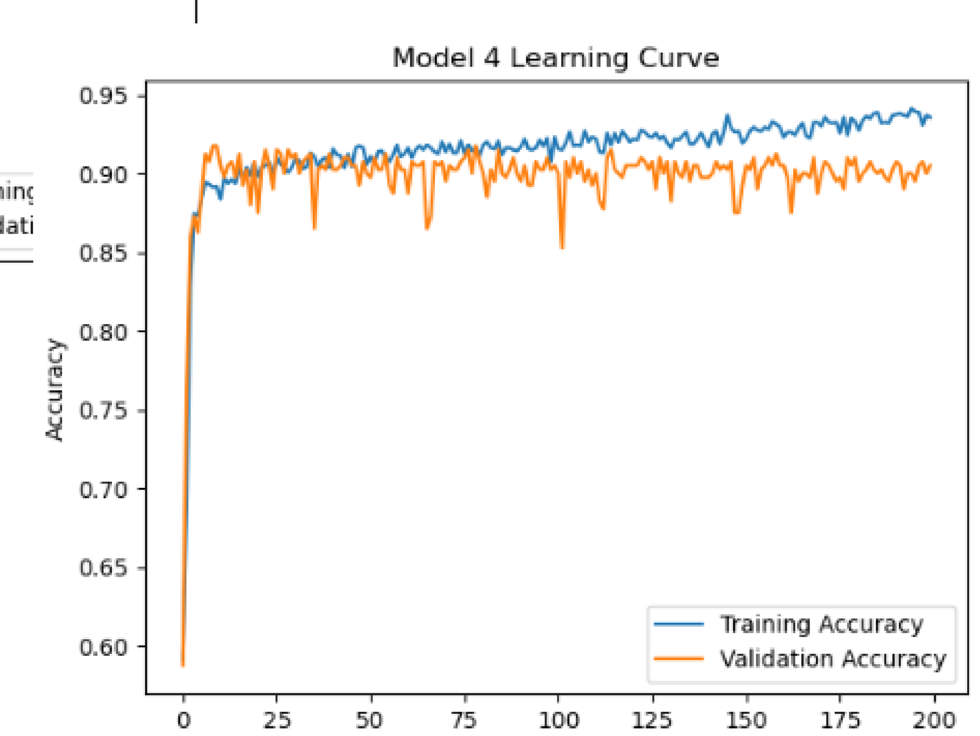
The models used:

- fully connected dense layers
- ReLU activation functions
- softmax output classification

Training and validation accuracy were monitored throughout training to study generalization performance.

	Model	Architecture	Validation Accuracy	Precision	Recall	F1 Score
0	Model 1	Output Only	0.8975	0.8974	0.8975	0.8942
1	Model 2	8 Hidden Neurons	0.9225	0.9219	0.9225	0.9215
2	Model 3	32 → 16 Hidden Neurons	0.9200	0.9196	0.9200	0.9192
3	Model 4	128 → 64 → 32 Hidden Neurons	0.9175	0.9181	0.9175	0.9167

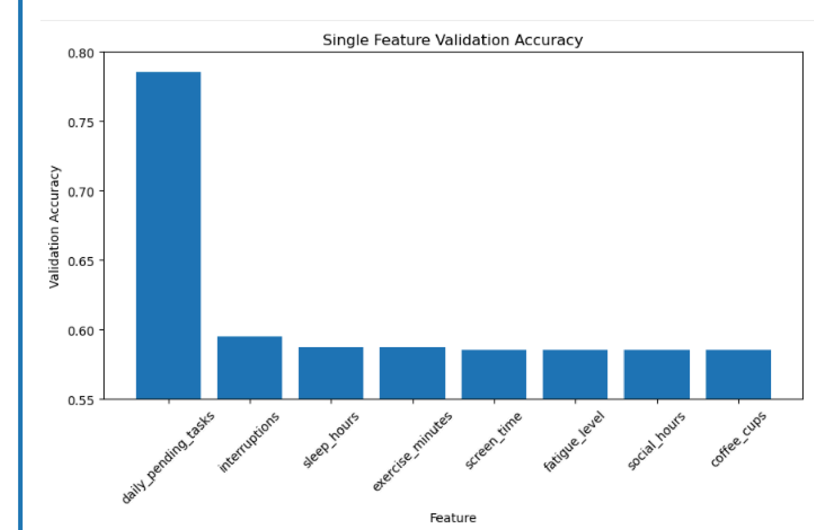


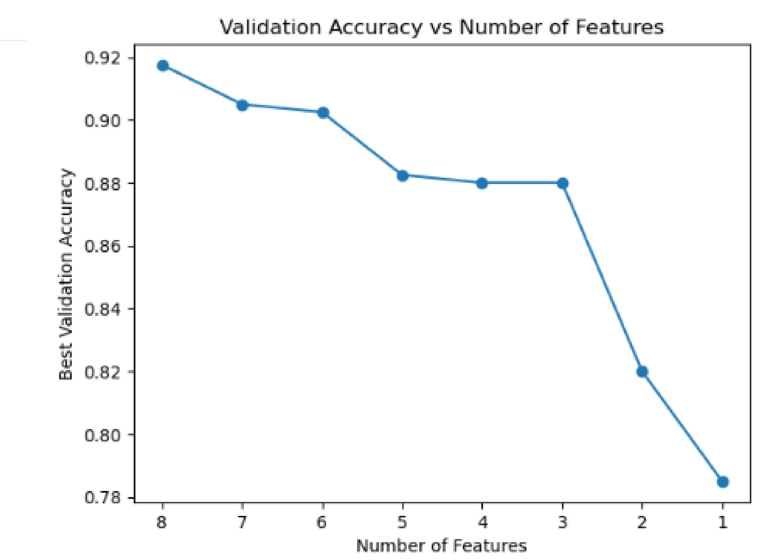


Feature Importance Analysis

Feature importance was analyzed by training models using one feature at a time. Validation accuracy was compared to determine which variables contributed most strongly to stress prediction.

Daily pending tasks showed the strongest predictive relationship with stress level classification.





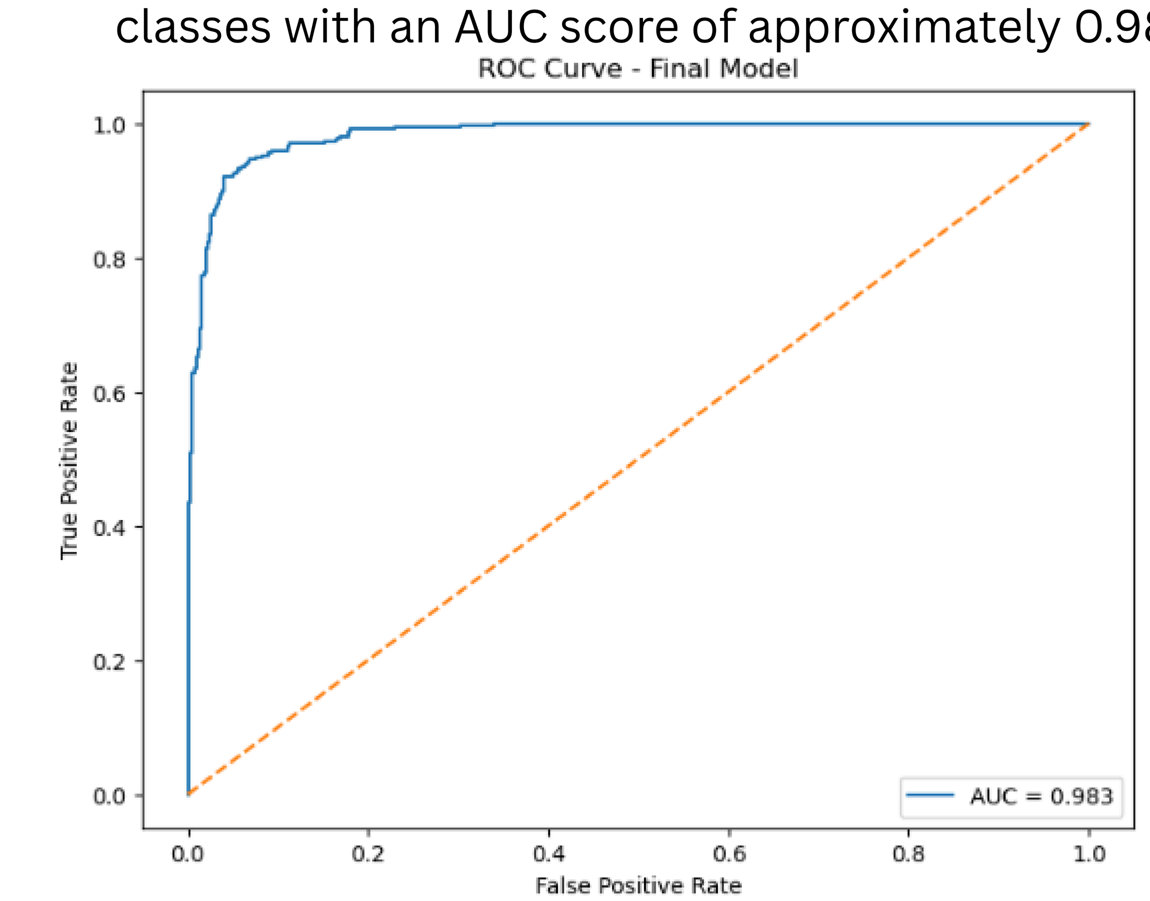
Final Model Evaluation

The final neural network model achieved strong classification performance on the validation dataset.

Performance was evaluated using:

- validation accuracy
- ROC analysis
- AUC score

The ROC curve demonstrated strong separation capability between classes with an AUC score of approximately 0.983.



AUC Score: 0.9834375

Limitations

- Synthetic dataset may not fully reflect real-world behavior
- Correlation does not imply causation
- Additional real-world healthcare data could improve generalization

Key Findings

- Hybrid baseline predictors outperformed simple additive approaches
- Daily pending tasks showed the strongest relationship with stress levels
- Neural networks achieved strong classification performance
- Increasing model complexity improved learning but also introduced overfitting risks
- Some lifestyle factors contributed very little to prediction performance